Bonus · Mathematics Challenge Test

40 questions · Calculator allowed · Use the Grade 8 Mathematics Reference Sheet

An extra full-length test for one more round of practice. Mark your answers on blank paper, then score with the key at the end.

Reporting Category 1 · Numerical Representations and Relationships

1. Order from **least to greatest**: ⅞, 0.9, 91%, √0.8. A. ⅞, √0.8, 0.9, 91% B. 91%, 0.9, √0.8, ⅞ C. 0.9, ⅞, 91%, √0.8 D. √0.8, ⅞, 0.9, 91%

2. Between which two whole numbers does √60 lie? A. 6 and 7 B. 7 and 8 C. 29 and 31 D. 59 and 61

3. *(Griddable)* Write 83,000 in scientific notation. ☐

4. Which number is **irrational**? A. √30 B. √36 C. ⅗ D. 0.2

Reporting Category 2 · Computations and Algebraic Relationships

5. What is the slope of the line through (2, 5) and (5, 14)? A. 2 B. 3 C. 4 D. 9

6. A proportional relationship passes through the origin and (8, 20). What is the unit rate? A. 2 B. 2.5 C. 4 D. 12

7. *(Equation editor)* A fund starts at $100 and grows $8 each week. Write an equation for the amount y after x weeks. ☐

8. Which relation is a **function**? A. {(1, 4), (1, 6), (2, 7)} B. {(1, 4), (2, 4), (3, 7)} C. {(5, 1), (5, 2), (6, 3)} D. {(0, 1), (0, 2), (1, 3)}

9. Which equation represents a **non-proportional** relationship? A. $y = 7x$ B. $y = \frac{2}{5}x$ C. $y = 4x - 3$ D. $y = 9x$

10. *(Griddable)* For a direct variation, $y = 42$ when $x = 6$. What is y when $x = 9$? ☐

11. Which table shows a **proportional** relationship? A. (1, 4)(2, 8)(3, 12) B. (1, 3)(2, 7)(3, 11) C. (1, 5)(2, 8)(3, 11) D. (2, 9)(3, 13)(4, 17)

12. *(Griddable)* Solve for x: $10x - 4 = 6x + 16$. ☐

13. "Four less than three times a number equals the number plus six." Which equation models this? A. $3x - 4 = x + 6$ B. $4 - 3x = x + 6$ C. $3x + 4 = x - 6$ D. $4x - 3 = x + 6$

14. Two lines intersect at (5, 2). What is the solution of the system? A. $x = 2, y = 5$ B. $x = 5, y = 2$ C. $x = 7$ D. No solution

15. On a line, the right triangles drawn to any two points are — A. congruent B. **similar** C. equilateral D. obtuse

16. Which situation is **proportional**? A. $20 plus $15 per hour B. 5 oranges for $2 at a steady rate C. 40 L draining 5 L/min D. $8 fee plus $3 each

17. Which situation can be modeled by $5x + 8 = 3x + 20$? A. Two savers start with $8 and $20, saving $5 and $3 per week; when equal? B. 5 times 8 C. A rectangle's area D. 8 plus 20

18. A florist charges $6 plus $2.50 per flower. Which equation gives the cost y for x flowers? A. $y = 6x + 2.5$ B. $y = 2.5x + 6$ C. $y = 8.5x$ D. $y = 2.5x - 6$

19. *(Select TWO)* Which statements about $y = 5x - 2$ are true? A. The slope is 5 B. The slope is −2 C. The y-intercept is −2 D. It passes through the origin

Reporting Category 3 · Geometry and Measurement

20. A cylinder has radius 2 and height 9. What is its volume? ($\pi \approx 3.14$) A. 56.52 B. 113.04 C. 226.08 D. 339.12

21. A cone has radius 6 and height 5. What is its volume? ($\pi \approx 3.14$) A. 94.2 B. 188.4 C. 376.8 D. 565.2

22. A cylinder has radius 3 and height 6. What is its **total** surface area? ($\pi \approx 3.14$) A. 113.04 B. 150.72 C. 169.56 D. 226.08

23. *(Griddable)* A right triangle has legs of 12 and 16. Find the hypotenuse. ☐

24. What is the distance between (−1, 2) and (2, 6)? A. 5 B. 7 C. 12 D. 25

25. The point (5, −2) is dilated by a scale factor of 3 about the origin. What is the image? A. (15, −6) B. (8, 1) C. (15, −2) D. (5, −6)

26. The point (−4, 9) is reflected over the x-axis. What is the image? A. (4, 9) B. (−4, −9) C. (4, −9) D. (9, −4)

27. The point (6, −3) is rotated 180° about the origin. What is the image? A. (−6, 3) B. (−3, 6) C. (3, −6) D. (6, 3)

28. An exterior angle of a triangle equals the sum of its two remote interior angles, 55° and 60°. What is the exterior angle? A. 65° B. 95° C. 115° D. 125°

29. Two similar figures have corresponding sides in a ratio of 2 : 3. A side of the smaller is 10. What is the matching side of the larger? A. 12 B. 13 C. 15 D. 20

30. A rectangle measures 8 by 2. It is dilated by a scale factor of 3. What is the area of the image? A. 16 B. 48 C. 96 D. 144

31. *(Griddable)* The base of a cylinder is a circle with radius 10. What is the base area B? ($\pi \approx 3.14$) ☐

32. Which transformation **changes** the size of a figure? A. Translation B. Reflection C. Rotation D. Dilation by factor 3

Reporting Category 4 · Data Analysis and Personal Financial Literacy

33. *(Griddable)* Find the mean absolute deviation of 3, 3, 9, 9. ☐

34. A trend line is modeled by y = 2.5x + 4. Predict *y* when x = 8. A. 20 B. 22 C. 24 D. 28

35. On a scatterplot, the points fall from upper-left to lower-right. What association is shown? A. Positive linear B. Negative linear C. No association D. Non-linear

36. Which pattern suggests the data are **linear**? A. A U-shaped curve B. A circular cluster C. A straight rising or falling path D. Random scatter

37. *(Griddable)* Find the simple interest on $2,500 at 4% for 3 years. ☐

38. For the same loan amount and length, how does a **higher** interest rate affect the total cost? A. Lowers it B. Raises it C. No effect D. Cancels the loan

39. What helps savings grow the most over time? A. Spending quickly B. Saving once C. Regular saving with compound interest D. Keeping cash at home

40. College costs $18,000. A family contributes $6,000 and saves the rest over 24 months. How much per month? A. $400 B. $500 C. $600 D. $750

Bonus Test — Answer Key & Explanations

1. A *(8.2D ★)* — ⅞=0.875, √0.8≈0.894, 0.9, 91%=0.91.

2. B *(8.2B ○)* — 49<60<64.

3. 8.3×10⁴ *(8.2C ○)* — 83,000.

4. A *(8.2A ○)* — √30 irrational; √36=6.

5. B *(8.4C ★)* — 9/3=3.

6. B *(8.4B ★)* — 20/8=2.5.

7. y = 8x + 100 *(8.5I ★)* — Start 100, rate 8.

8. B *(8.5G ★)* — Each input once.

9. C *(8.5B ○)* — −3 → non-proportional.

10. 63 *(8.5E ○)* — k=7; 7×9.

11. A *(8.5F ○)* — Ratio 4 constant.

12. 5 *(8.8C ★)* — 4x=20.

13. A *(8.8A ○)* — 3x−4=x+6.

14. B *(8.9A ○)* — (5,2).

15. B *(8.4A ○)* — Similar triangles.

16. B *(8.5H ○)* — Steady rate, no fee.

17. A *(8.8B ○)* — 5x+8=3x+20.

18. B *(8.5B ○)* — y=2.5x+6.

19. A, C *(8.5I ★)* — Slope 5, y-int −2.

20. B *(8.7A ★)* — 3.14×4×9=113.04.

21. B *(8.7A ★)* — ⅓×3.14×36×5=188.4.

22. C *(8.7B ★)* — 36π+18π=54π≈169.56.

23. 20 *(8.7C ★)* — 144+256=400.

24. A *(8.7D ○)* — Legs 3,4→5.

25. A *(8.3C ★)* — (15,−6).

26. B *(8.10C ★)* — Over x-axis → (−4,−9).

27. A *(8.10C ★)* — 180° → (−6,3).

28. C *(8.8D ○)* — 55+60=115°.

29. C *(8.3A ○)* — 10×3/2=15.

30. D *(8.10D ○)* — Area ×9: 16×9=144.

31. 314 *(8.6A ○)* — 3.14×100.

32. D *(8.10A/B ○)* — Dilation changes size.

33. 3 *(8.11B ○)* — Mean 6; all distances 3.

34. C *(8.5D ★)* — 2.5×8+4=24.

35. B *(8.11A ○)* — Falling → negative linear.

36. C *(8.5C ○)* — Straight path → linear.

37. $300 *(8.12D ★)* — 2,500×0.04×3.

38. B *(8.12A ○)* — Higher rate raises cost.

39. C *(8.12C ○)* — Regular + compound grows most.

40. B *(8.12G ○)* — (18,000−6,000)/24=500.